

Project Charter

Date: 7th March 2026

Team Name: P1-A25

★ Executive Summary

To develop a data-driven diagnostic support system to assist in the early identification of Parkinson’s Disease using EEG signals. By applying signal processing techniques using the MNE-Python framework alongside deep learning models such as EEGNet and traditional machine learning methods, the system will analyse both resting-state and task-based brain activity to distinguish between individuals with Parkinson’s Disease and healthy controls. The final outcome will be a Streamlit-based application designed to provide clinicians with objective, data-informed insights to support diagnostic decision-making.

🎯 Project Objectives (SMART)

- Develop and validate an EEG-based diagnostic support prototype for Parkinson’s Disease using the EEG: 3-Stim Auditory Oddball and Rest in Parkinson’s dataset.
- Achieve performance aligned with published EEG-based detection benchmarks, targeting ≥80% classification accuracy and ≥80% sensitivity.
- Implement and compare EEGNet with selected traditional machine learning models to evaluate performance and robustness.
- Validate results using stratified 5-fold cross-validation to reduce overfitting and assess generalisability.

🚩 Project Scope

- In-Scope:**
- EEG preprocessing (filtering, epoching, artefact removal, feature extraction) using MNE.
 - Model design, training, and validation (EEGNet and traditional ML) using stratified 5-fold cross-validation.
 - Development of a Streamlit-based prototype application.
 - Integration of an ethical toolkit to support responsible AI practices.
- Out-of-Scope:**
- Regulatory approval processes.
 - EEG hardware development.
 - Clinical trials

👤 Team & Stakeholders

- Project Team**
- Justyna Dobersztajn – Project Manager & Business Analyst
 - Diba Dabiransari – Software Developer & Software Architect
 - Aleeza Azad – DevOps Engineer & Technical Writer
- Key Stakeholders**
- Dr. Ali Elmahmudi – Client
 - Dr. Kulvinder Panesar – Supervisor / Module Leader
- Additional Stakeholders**
- AI Ethics Specialist
 - Clinical Neurologist

📋 Project Deliverables

- Cleaned and preprocessed EEG dataset using MNE, including filtering, artefact removal, epoching, and feature selection.
- Trained and validated classification models (EEGNet and traditional ML) for Parkinson’s detection.
- Streamlit-based prototype accepting EEG input and generating prediction outputs with key performance metrics.
- Technical documentation outlining system design, methodology, validation, evaluation, and implementation.

💬 Considerations (Ethical & Legal)

- **GDPR Compliance:** Use of anonymised, publicly available EEG data in accordance with data protection regulations.
- **Ethical Toolkit Integration:** Application of responsible AI practices, including bias mitigation and transparency measures.
- **Model Explainability:** Clear and interpretable classification outputs to enhance understanding of predictions.
- **Human Oversight:** The system is designed to support, not replace, clinical judgement.
- **Prototype Deployment Only:** Limited to academic demonstration purposes and not intended for real-world clinical use yet.

 **Success Criteria**

- Achieve ≥80% accuracy and ≥80% sensitivity, aligned with published benchmarks.
- Validate performance using stratified 5-fold cross-validation.
- Support evaluation with a confusion matrix and key classification metrics.
- Demonstrate clear EEG signal visualisation before and after preprocessing to evidence data quality improvement.
- Deliver a functional Streamlit-based prototype by April 2026.

 **Budget Overview**

- **Team:** £35,000–£40,000 (based on industry salary benchmarks for AI engineering roles)
- **Computing Resources:** Google Colab Pro and cloud storage (£300–£600).
- **AI & Development Tools:** ChatGPT Plus, Gemini, GitHub Copilot, Visual Studio licences (£400–£600).
- **Project Management & Design Tools:** Jira, Trello, GitHub (mostly free), Canva Pro (£13 per month).

 **Risks & Challenges**

- **Preprocessing Sensitivity:** Inaccurate filtering or artefact removal may distort signal features and impact model performance.
- **Model Complexity & Overfitting:** Selecting an overly complex model may reduce generalisability and complicate interpretation.
- **Deployment Limitations:** Technical issues during Streamlit integration or environment configuration may impact prototype stability and performance.
- **Ethical & Privacy Considerations:** Potential misinterpretation of results, over-reliance on automated predictions, and privacy risks when handling patient-related data.

 Project Timeline	
Project Initiation	October 2025 - November 2025
Sprint 1 - Research and Planning	November 2025 - January 2026
Sprint 2 - Data Preprocessing	January 2026 - February 2026
Sprint 3 - Modelling and Evaluation	February 2026 - Mid March 2026
Sprint 4 - Deployment	Mid March 2026 - April 2026
Scrum Closure	Mid April 2026